

Operon Logic: predict the mutant

Two inputs, one promoter, and four regulatory mutants that can only be told apart by reasoning about mechanism rather than by memorising a phrase.

Name _____

Date _____

Period _____

Demonstration: andresdbp.github.io/resources/demos/operon-logic/

1 Before you open anything

Answer from what you already know. Whatever you write here gets compared against a number later, so commit to something specific.

PREDICTION

Wild-type *E. coli* growing in medium that contains **lactose and glucose together**. Taking full induction (lactose present, glucose absent) as 100%, roughly what percentage of maximum is *lacZYA* being transcribed at?

Write one or two sentences saying what reasoning got you that number.

2 Reading the operon in all four conditions

Open the demonstration on the **lac operon** tab. The *Genotype* menu should read **Wild type**. The four buttons under *Conditions in the medium* set the sugars; the *Output* panel reports three things at once.

1. Press the preset **lac- glu-**.
2. Copy three readouts into the first row of the table: **Transcription of lacZ Y A** (the large percentage), **Repressor on operator** and **CAP-cAMP on CAP site**.
3. Work down the remaining three presets and fill in the rest of the table.

Condition	Transcription (% of max)	Repressor on operator	CAP-cAMP on CAP site
lac- glu-			
lac- glu+			
lac+ glu-			
lac+ glu+			

a. Exactly one row has the repressor *off* the operator and CAP-cAMP *on* its site at the same time. Which row, and what does the transcription readout say there?

b. Compare the last two rows. One of the two protein readouts is identical in both; the other flips. Name the readout that flips and name the small molecule whose concentration is responsible for it.

c. The two lactose-free rows do not read zero. Using the *Honest limits* panel, state in one sentence why a completely silent operon would be a problem for the cell.

The inducer is **allolactose**, an isomer that β -galactosidase makes from lactose — not lactose itself. The map draws them as one amber ligand for clarity. Say “allolactose binds the repressor” on paper.

3 What the glucose switch is actually doing

Here you change one control, the glucose preset, and hold everything else fixed while stepping through the whole *Genotype* menu. Lactose stays present throughout.

1. Set *Genotype* to **Wild type**.
2. Press **lac+ glu-** and record the transcription percentage. Press **lac+ glu+** and record it again.
3. Divide the first number by the second and write the ratio in the last column.
4. Repeat for each of the other four genotypes in the menu.

Genotype	lac+ glu- (%)	lac+ glu+ (%)	Ratio (glu- ÷ glu+)
Wild type			
lacI ⁻			
O ^c			
lacI ^s			
CAP ⁻			

a. Four of the five ratios are the same to within rounding. Write that ratio, and complete the rule: adding glucose divides transcription by roughly _____, and it does so whether the operon was sitting at 100% or at 0.5%.

b. One genotype has a ratio of 1 — glucose changes nothing at all. Name it, and explain in one or two sentences why that single result rules out the idea that glucose blocks the operator.

c. Go back to your prediction in Part 1. Compare it with the wild-type **lac+ glu+** value. If it was off, say specifically what you assumed about the inputs that turned out to be incomplete.

4 Using it the way a question would

Exam questions rarely hand you the genotype and ask for the number. They hand you the numbers and ask for the genotype, or they hand you a plausible shortcut and see whether you check it.

A. Predict before you look.

1. Set *Genotype* to O^c first, then press **Start prediction round**. The transcription numbers hide; the molecular map keeps working.
2. Reading the map alone, choose OFF, LOW or HIGH for each of the four conditions and write your four choices below *before* you press **Reveal**.
3. Press **Reveal** and record the actual value beside each.

Condition	Your prediction	Actual (% of max)	OFF / LOW / HIGH
no lactose, no glucose			
no lactose, glucose present			
lactose present, no glucose			
lactose AND glucose present			

a. Score recorded by the demonstration: _____ / 4. For any box you missed, write what the map was telling you that you did not use.

B. Working backwards from a pattern.

b. A strain reads **OFF, OFF, LOW, LOW** across the four conditions in that order, and the *Repressor on operator* readout still responds normally to lactose. Which genotype in the menu is it? Select it and confirm, then write the two features of the pattern that identify it.

C. A shortcut worth testing.

"Once the repressor is gone, nothing is holding the operon back, so it should run at maximum." That is a reasonable line of thought. Test it.

1. Set *Genotype* to **lacI⁻** and press **lac- glu⁻**. Record the percentage and the *Repressor on operator* readout.
2. Press **lac- glu⁺**. Record both again.

Condition	Transcription (% of max)	Repressor on operator
lac- glu-		
lac- glu+		

c. The repressor readout is the same in both rows, yet the transcription level is not. Explain in two sentences why losing the repressor does not make the operon glucose-insensitive.

Negative regulation and positive regulation are separate events on the same promoter. Removing a blocker is not the same thing as supplying a helper, and a strain can lose one while keeping the other intact.

5 Solve these without the demonstration

Close the page, or look away from it. Show your reasoning, not just the answer. Then reopen it and check — and where you were wrong, write down which step in the mechanism you skipped.

1. A strain transcribes *lacZYA* at 100% of maximum in medium with **no lactose and no glucose**, and at about 4.9% in medium with **no lactose but glucose present**. Name the two genotypes from the menu that both fit these numbers, and state the experiment that separates them and what the result would be for each.

2. A CAP⁻ strain is grown in lactose with no glucose. Give the transcription level as a percentage of the wild-type maximum, and explain why the number is neither 0% nor 100%.

3. In a *lacI^s* strain, glucose is added to a medium that already contains lactose. By what factor does transcription change? Predict the factor and justify it before you check — and say why it is the same factor as in wild type even though the absolute levels differ by more than a hundredfold.

4. Explanation only, no numbers. Explain why deleting the repressor leaves the operon fully glucose-sensitive, while deleting CAP makes the glucose signal disappear entirely. Your answer should name what each protein does to the promoter and say which one glucose acts through.

6 On your own

Nothing here is graded. The second tab, **Eukaryotic enhancers**, is untouched by everything above, and it is where this unit is going.

- Switch to *Eukaryotic enhancers* and cycle the *Which cell are you in?* menu through hepatocyte, skeletal muscle cell and neuron. Record all three transcription levels, then use the *Silencer occupied* readout to say why muscle and neuron are both “off” for different reasons — and why the neuron sits lower.
- Select the neuron, then tick *HNF-4α* and *C/EBPα*. The gene barely moves. Now untick *MeCP2*. What does the order of those two results tell you about a repressor sitting on top of a set of activators? Press *Back to the real cell* when you are done.
- Starting from Sp1 alone, add HNF-4α on its own and note the level; go back and add C/EBPα on its own and note that level; then have both bound at once. Add the two separate increases together and compare with the level for both. Then select the skeletal muscle cell and read the tag beside the *MyoD* checkbox — it states why an abundant and genuinely powerful activator does nothing to this particular gene.